



POSEIDON LASER BEAM XS

MANUAL

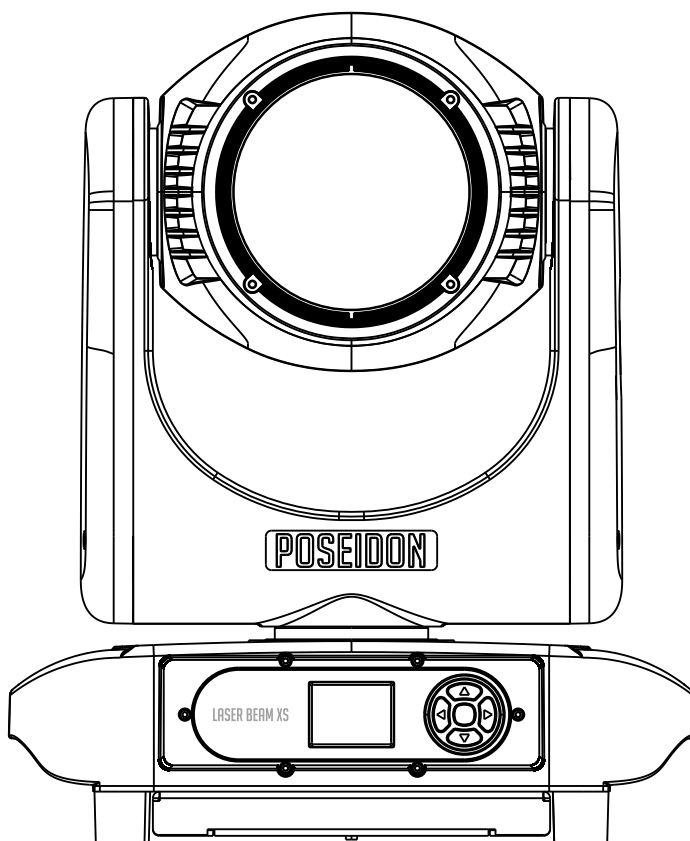
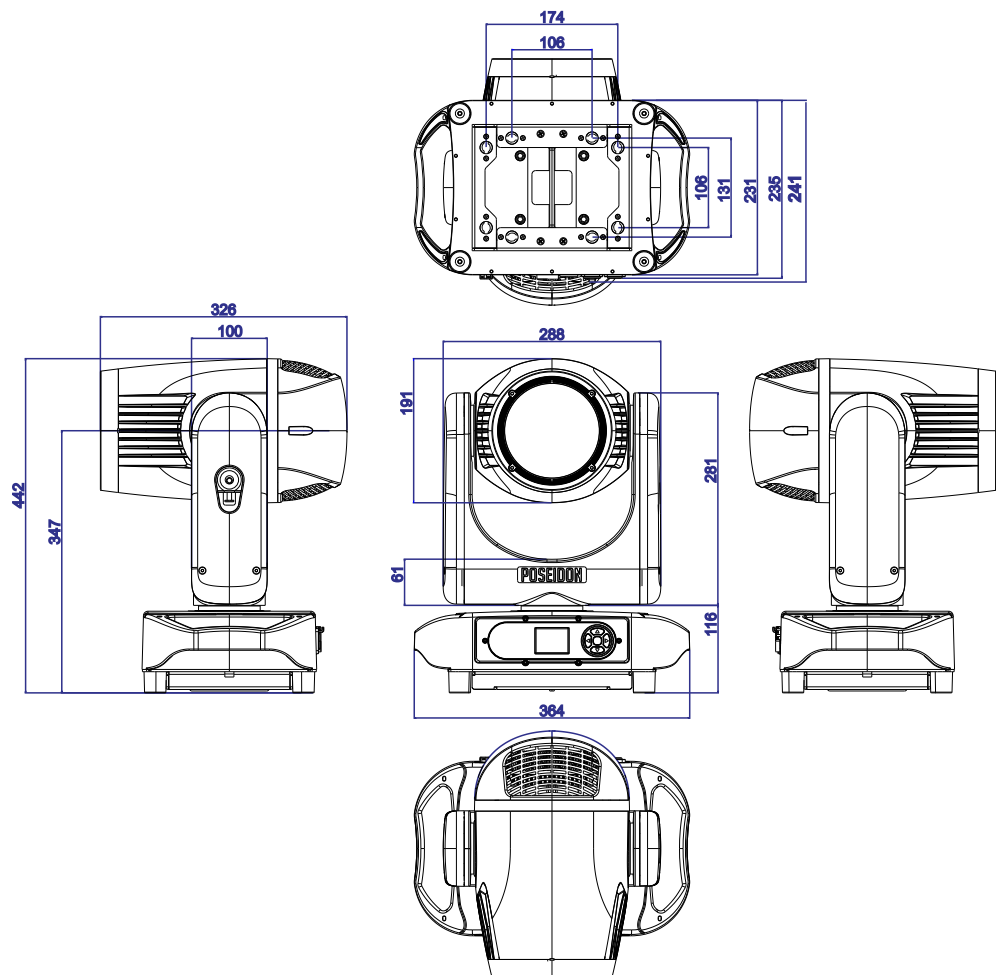


TABLE OF CONTENTS

Dimensions	1
Safety instruction	2
Protection from burns and fire	3
Protection from injury	3
Photobiological safety	4
Fixture overview	5
Introduction	6
Powerful outdoor laser beam xs	6
Using for the first time	6
AC Power	6
Power voltage	6
Power cables	7
Relaying power to other devices	7
Data link	7
Tips for reliable data transmission	7
Physical installation	8
Fastening the fixture to a flat surface	8
Rigging via omega bracket	9
Outdoor IP-rated fixtures	10
Condensation/moisture inside housing	10
Fixtures temperature specification	10
Temporary usage	10
Setup	11
Control panel and menu navigation	11
DMX address setting	11
Onboard control menu	12
DMX protocol	16
Gobo overview	19
Photometrics	20
Circuit connection diagram	21
Specifications	22
Positioning	24

DIMENSIONS

ALL DIMENSIONS ARE IN MILLIMETERS



SAFETY INSTRUCTION



WARNING!
Read the safety precautions in this section
before installing, powering, operating or
servicing this product.

The following symbols are used to identify important safety information on the product and in this manual:



DANGER!
Safety hazard.
Risk of severe
injury or death.



DANGER!
Hazardous
voltage. Risk of
lethal or severe
electric shock.



WARNING!
Fire hazard.



WARNING!
LED light
emission. Risk of
eye injury.



WARNING!
Burn hazard. Hot
surface. Do not
touch.



WARNING!
Wear protective
eyewear.



WARNING!
Refer to user
manual.



Do not look into the beam at short distance of the of the product.
Do not view the light output with optical instruments or any device that may concentrate the beam.



This product is for professional use only. It is not for household use.
This product presents risks of severe injury or death due to fire and burn hazards, electric shock and falls.
Read this manual before installing, powering or servicing the fixture, follow the safety precautions listed below and observe all warnings in this manual and printed on the fixture. If you have questions about how to operate the fixture safely, please contact your supplier.



PROTECTION FROM ELECTRIC SHOCK



- Disconnect the fixture from AC power before removing or installing any cover or part.
- Always ground (earth) the fixture electrically.
- Use only a source of AC power that complies with local building and electrical codes and has both overload and ground-fault (earth-fault) protection.
- Before using the fixture, check that all power distribution equipment and cables are in perfect condition and rated for the current requirements of all connected devices.
- Power input and throughput cables must be rated 20A minimum, have three conductors 1.5 mm² (16 AWG) minimum conductor size and an outer cable diameter of 5 - 15 mm. Cables must be hard usage type (SJT or equivalent) and heat-resistant to 90°C minimum.
- Use only PowerCON TRUE 1® cable connectors to connect to power input sockets. Use only PowerCON TRUE1® cable connectors to connect to power throughput sockets.
- Isolate the fixture from power immediately if the power plug or any seal, cover, cable, or other component is damaged, defective, deformed, wet or showing signs of overheating. Do not reapply power until repairs have been completed.
- Refer any service operation not described in this manual to a qualified technician.
- Socket outlets used to supply fixture fixtures with power or external power switches must be located near the fixtures and easily accessible so that the fixtures can easily be disconnected from power.

PROTECTION FROM BURNS AND FIRE



- The exterior of the fixture becomes hot during use. Avoid contact by persons and materials. Allow the fixture to cool for at least 5 minutes before handling.



- Keep all combustible materials (e.g. fabric, wood, paper) at least 1 metres away from the fixture.
- Keep flammable materials well away from the fixture.
- Ensure that there is free and unobstructed airflow around the fixture.
- Do not illuminate surfaces within 12 metres of the fixture.
- Do not attempt to bypass thermostatic switches or fuses.
- If you relay power from one fixture to another using power throughout sockets, do not connect more than five fixtures in total to each other in an interconnected chain.
- Connect only other fixtures to fixture power throughout sockets.
- Do not stick filters, masks or other materials onto any optical component.
- Do not modify the fixture in any way not described in this manual.

PROTECTION FROM INJURY



- Fasten the fixture securely to a fixed surface or structure when in use. The fixture is not portable when installed.
- Ensure that any supporting structure and/or hardware used can hold at least 10 times the weight of all the devices they support.



- Allow enough clearance around the head to ensure that it cannot collide with an object or another fixture when it moves.
- Check that all external covers and rigging hardware are securely fastened.
- Block access below the work area and work from a stable platform whenever installing, servicing or moving the fixture.
- Do not operate the fixture with missing or damaged covers, shields or any optical component.

PHOTOBIOLOGICAL SAFETY



- The light source of this product is based on laser diodes.
- This product qualifies for the laser products safety standard IEC 60825-1:2014, edition 3, "part 4.4, Laser products designed to function as conventional lamps", under which it is classified as CLASS 1 LASER PRODUCT. Alternately evaluated under the standard IEC 62471-5:2015 "Photobiological safety of lamps and lamp systems", the photobiological risk classification is assigned as RISK GROUP 3 (RG3).



- **CAUTION!** Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.



Caution! Possibly hazardous optical radiation emitted from this product. do not look at operating lamp source.
Eye injury may result.



RG3
Hazard distance: Refer to the manual.
Not for household use.
EN/IEC 62471-5



CLASS 1 LASER PRODUCT
EN/IEC 60825-1 2014



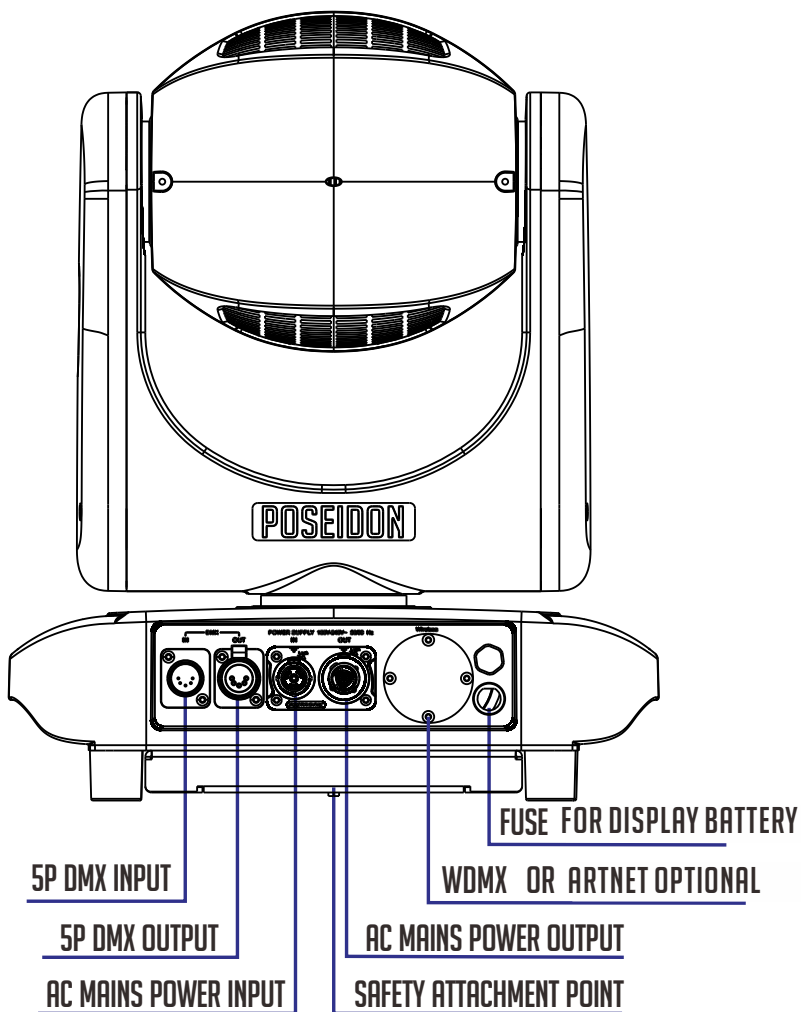
RISK GROUP 3
WARNING! DO NOT STARE INTO THE BEAM.
EYE EXPOSURE TO THE BEAM IS NOT PERMITTED.
EN/IEC 62471
HAZARD DISTANCE: REFER TO THE MANUAL.

- No direct exposure to the beam shall be permitted, RG3 IEC 62471 -5:2015.
- Operators shall control access to the beam within the hazard distance or install the product at the height that will prevent spectators eyes from being in the hazard distance.
- According to IEC 62471-5, the distance from RG3 to RG1 is 45 meters.
- **CAUTION!** Use of controls or adjustments or performance of procedures other than those specified herein may result in hazardous radiation exposure.
- The laser safety classification of this product is based on IEC 60825-1: 2014.
- The laser module information is as follows :



Wavelength:	Main wavelength 455nm
Beam divergence:	50°
Maximum output power	120W
Laser safety level:	CLASS 4

FIXTURE OVERVIEW



INTRODUCTION

POWERFUL OUTDOOR LASER BEAM XS

- 120W LASER SOURCE
- 128MM LENS FOR 1.1° BEAM
- CMY AND COLOR WHEEL
- GOBO WHEEL, FROST AND 2 PRISMS
- LIGHT WEIGHT SLIM HOUSING
- IP65 DUST AND WATER PROTECTION

USING FOR THE FIRST TIME



Warning! Read “Safety Information” before installing, powering, operating or servicing the fixture.
Before applying power to the fixture:

Check that the local AC mains power source is within the fixture's power voltage and frequency ranges.

See “Power cables and power plug” on page 6. Install a PowerCON TRUE1 ® power input connector power cable.

AC POWER



Warning! Read “Safety Information” starting on before connecting the fixtures to AC mains power.

Warning! For protection from electric shock, the fixture must be grounded (earthed). The power distribution circuit must be equipped with a fuse or circuit breaker and ground-fault (earth-fault) protection.

Warning! Socket outlets or external power switches used to supply the fixture with power must be located near the fixture and easily accessible so that the fixtures can easily be disconnected from power.



Important! Do not insert or remove live PowerCON TRUE 1 ® connectors to apply or cut power, as this may cause arcing at the terminals that will damage the connectors.

Important! Do not use an external dimming system to supply power to the fixture, as this may cause damage to the fixture that is not covered by the product warranty.

The fixture can be hard wired to a electrical installation if you want to install it permanently, or a power plug that is suitable for the local power outlets can be installed on the power cable.

POWER VOLTAGE



Warning! Check that the voltage range specified on the fixture serial number label matches the local AC mains power voltage before applying power to the fixture.

The fixtures accepts AC mains power at 100-240V nominal, 50/60 Hz. Do not apply AC mains power to the fixture at any other voltage than specified.



Warning! This product contain a ni-mh battery

POWER CABLES

Power input and throughput cables must be rated 16A minimum, have three conductors 1.5 mm² (16 AWG) minimum conductor size and an outer cable diameter of 5 - 15 mm. Cables must be hard usage type (SJT or equivalent) and heat- resistant to 90°C minimum. In the EU the cable must be HAR approved or equivalent.

If you install a power plug on the power cable, install a grounding-type (earthed) plug that is rated 16A minimum. Follow the plug manufacturer's instructions. Table 1 shows standard wire color-coding schemes and some possible pin identification schemes; if pins are not clearly identified.

Wire Color (EU models)	Wire Color (US models)	Conductor	Symbol
Brown	Black	Live	L
Blue	White	Neutral	N
Yellow/Green	Green	Ground (earth)	 or 

Table 1: Wire color-coding and power connections

RELAYING POWER TO OTHER DEVICES

Warning! Do not connect more than four fixtures in total in one interconnected chain. Power can be relayed to another device via the PowerCON TRUE 1 ® throughput socket.

If you daisy chain the fixtures in a chain so that they all draw AC mains power via the first fixture, certain points must be respected:
A heavy duty, three-conductor, 16 AWG or 1.5 mm² cable with SJT or equivalent cable jacket must be used to connect the first fixture to AC mains power. PowerCON TRUE1 ® connectors must be used to draw AC mains power from the fixtures power throughput socket and yellow PowerCON TRUE 1 ® connectors must be used to supply power at the fixture's power input sockets.

DATA LINK

A DMX 512 data link is required in order to control a fixture via DMX. The fixture has 5-pin XLR connectors for DMX data input and output.
The pin-out on all connectors is pin 1 = shield, pin 2 = cold (-), and pin 3 = hot (+) Pins 4 and 5 in the 5-pin XLR connectors are not in use.

TIPS FOR RELIABLE DATA TRANSMISSION

To connect the fixture to data:

1. Connect the DMX data output from the controller to the 5-pin XLR connector of the nearest fixture.
2. Connect the DMX output of the first fixture to the DMX input of the next fixture and continue connecting fixtures.

PHYSICAL INSTALLATION

Warning! The fixture must be either fastened to a flat surface such as a stage or wall, or clamped to a truss or similar structure in any orientation using a rigging clamp.



Warning! Always attach an approved safety cable to one of the safety cable attachment points on the base.

Do not illuminate surfaces within 12 meters of the fixture. Ensure that flammable materials (wood, fabric, paper, etc.) are minimum 1 meters from the fixture and allow a free airflow around the fixture.



FASTENING THE FIXTURE TO A FLAT SURFACE

The fixture can be fastened to a fixed flat surface that is oriented at any angle. Check that the surface can support at least 10 times the weight of all fixtures and equipment to be installed.

Warning! The supporting surface must be hard and flat or cooling may be blocked, which will cause overheating. Fasten the fixture securely. Do not place it on unstable surfaces. Always attach a securely anchored safety cable to the safety cable attachment point.

1. Block access under the work area. Working from a stable platform, hang the fixture on the truss with the arrow on the base towards the area to be illuminated. Tighten the rigging clamp.
2. Secure the fixture against clamp failure with a secondary attachment such as an approved safety cable that is rated for the weight of the fixture using one of the attachment points at the edges of the base (see "Fixture overview"). Do not use any other part of the fixture as a safety cable attachment point.

The installation of the fixture has to be built and constructed in a way that it can hold 10 times the weight for 1 hour without any harming deformation.

The installation must always be secured with a secondary safety attachment, e.g. an appropriate catch net. This secondary safety attachment must be constructed in a way that no part of the installation can fall down if the main attachment fails.

When rigging, derigging or servicing the fixture staying in the area below the installation place, on bridges, under high working places and other endangered areas is forbidden.

The operator has to make sure that safety-relating and machine-technical installations are approved by an expert before taking into operation for the first time and after changes before taking into operation another time.

The operator has to make sure that safety-relating and machine-technical installations are approved by an expert after every four year in the course of an acceptance test.

The operator has to make sure that safety-relating and machine-technical installations are approved by a skilled person once a year.

The fixture should be installed outside areas where persons may walk by or be seated.

IMPORTANT! OVERHEAD RIGGING REQUIRES EXTENSIVE EXPERIENCE, including (but not limited to) calculating working load limits, installation material being used, and periodic safety inspection of all installation material and the projector. If you lack these qualifications, do not attempt the installation yourself, but instead use a professional structural rigger. Improper installation can result in bodily injury or damage to property. The fixture has to be installed out of the reach of people.



If the fixture shall be lowered from the ceiling or high joists, professional trussing systems have to be used. The fixture must never be fixed swinging freely in the room.

Caution: Fixture may cause severe injuries when crashing down! If you have doubts concerning the safety of a possible installation, do not install the moving head!

Before rigging make sure that the installation area can hold a minimum point load of 10 times the fixture's weight.

When installing the device, make sure there is no highly inflammable material (decoration articles, etc.) in a distance of min. 1m.

Use an appropriate clamp to rig the fixture on the truss.

Follow the instructions mentioned at the bottom of the base.

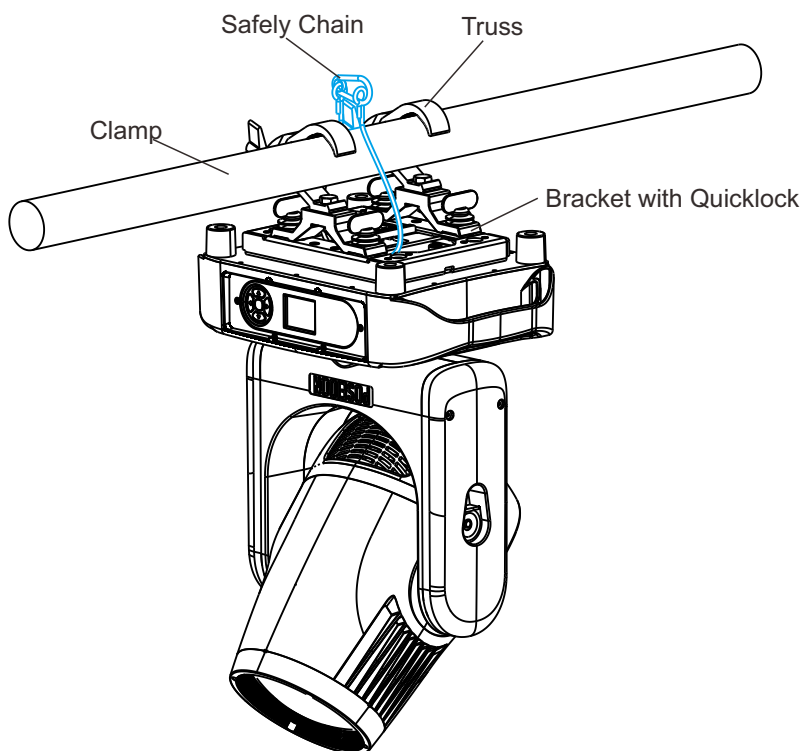
Make sure that the device is fixed properly! Ensure that the structure (truss) to which you are attaching the fixtures is secure.



The fixture can be placed directly on the stage floor or rigged on a truss without altering its operation characteristics .

For securing the fixture to the truss, install a safety wire which can hold at least 10 times the weight of the fixture. Use only the safety wire with a snap hook with screw lock gate. Fasten the safety cable in the attachment point and around the truss as shown on the picture below.

RIGGING VIA OMEGA BRACKET:



OUTDOOR IP-RATED FIXTURES

CLF products are applied to official classified IP norm levels. For this product the IP rate is IP65 when using the covers for the chassis parts. IP65 means according classified norm: shielded against dust and pressurized water from any side. Typical use for outdoor rated stage events with normal weather acceptance. So no heavy rain, because then the water pressure over exceeds the IP norm.

SOLID OBJECT

1	Protected against a solid object greater than 50mm such as a hand.
2	Protected against a solid object greater than 12.5mm such as a finger.
3	Protected against a solid object greater than 2.5mm such as a screwdriver.
4	Protected against a solid object greater than 1mm such as a wire.
5	Dust protected. Limited ingress of dust permitted. Will not interfere with operation of the equipment.
6	Dust tight. No ingress of dust.

IP 65

Ingress Protection

MOISTURE

1	Protected against vertical falling drops of water. Limited ingress permitted.
2	Protected against vertical falling drops of water with enclosure tilted up to 15 degrees from the vertical. Limited ingress permitted.
3	Protected against sprays of water up to 60 degrees from the vertical. Limited ingress permitted.
4	Protected against water splashes from all directions. Limited ingress permitted.
5	Protected against jets of water. Limited ingress permitted.
6	Protected against powerful jets of water. Limited ingress permitted.
7	Protected against the effects of immersion in water between 15cm and 1m for 30 minutes.
8	Protected against the effects of immersion in water under pressure for long periods.

CONDENSATION/MOISTURE INSIDE HOUSING

Because of high humidity levels during production condensation can occur inside the housing. This is mostly visible on the coldest parts of the fixture, like the front glass or display. To prevent this problem we work with special conditioned areas for outdoor fixtures. Because of the breathing air valves it is still possible to get humidity inside the fixture. This will evaporate slowly. Do not put wet fixtures in a flightcase, this will help humidity enter the fixture.

FIXTURES TEMPERATURE SPECIFICATION

Make sure the fixture is used within its working temperature range. Outside this range we cannot guarantee correct operation.

TEMPORARY USAGE:

Stage event equipment is designed for temporary outdoor use. Materials are not designed for long-term exposure to heavy weather conditions. Rubber covers will be negatively affected by long-term UV exposure and should be checked by qualified service technicians over time. Tightening screws too hard will negatively affect the IP-rating.

SETUP

Warning! Read "Safety Information" before installing, powering, operating the fixture.

CONTROL PANEL AND MENU NAVIGATION

The onboard control panel and backlit graphic display are used to adjust the DMX address, fixture settings (personality), service utilities. See "Onboard control menus" for a complete list of menus and commands.

Using the control buttons:

- To enter the menu select [ENTER].
- Press [UP], [DOWN], [LEFT] AND [RIGHT] to scroll within a menu or adjust values.
- To enter a menu, select a function or apply a selection, press [ENTER].
- To escape a function or move back one level in the menu structure, press [LEFT].

DMX ADDRESS SETTING

The DMX address is the first channel used to receive instructions from the controller. For independent control, each fixture must be assigned to a separate channel. The DMX address can be configured by using the DMX ADDRESS menu in the control panel.

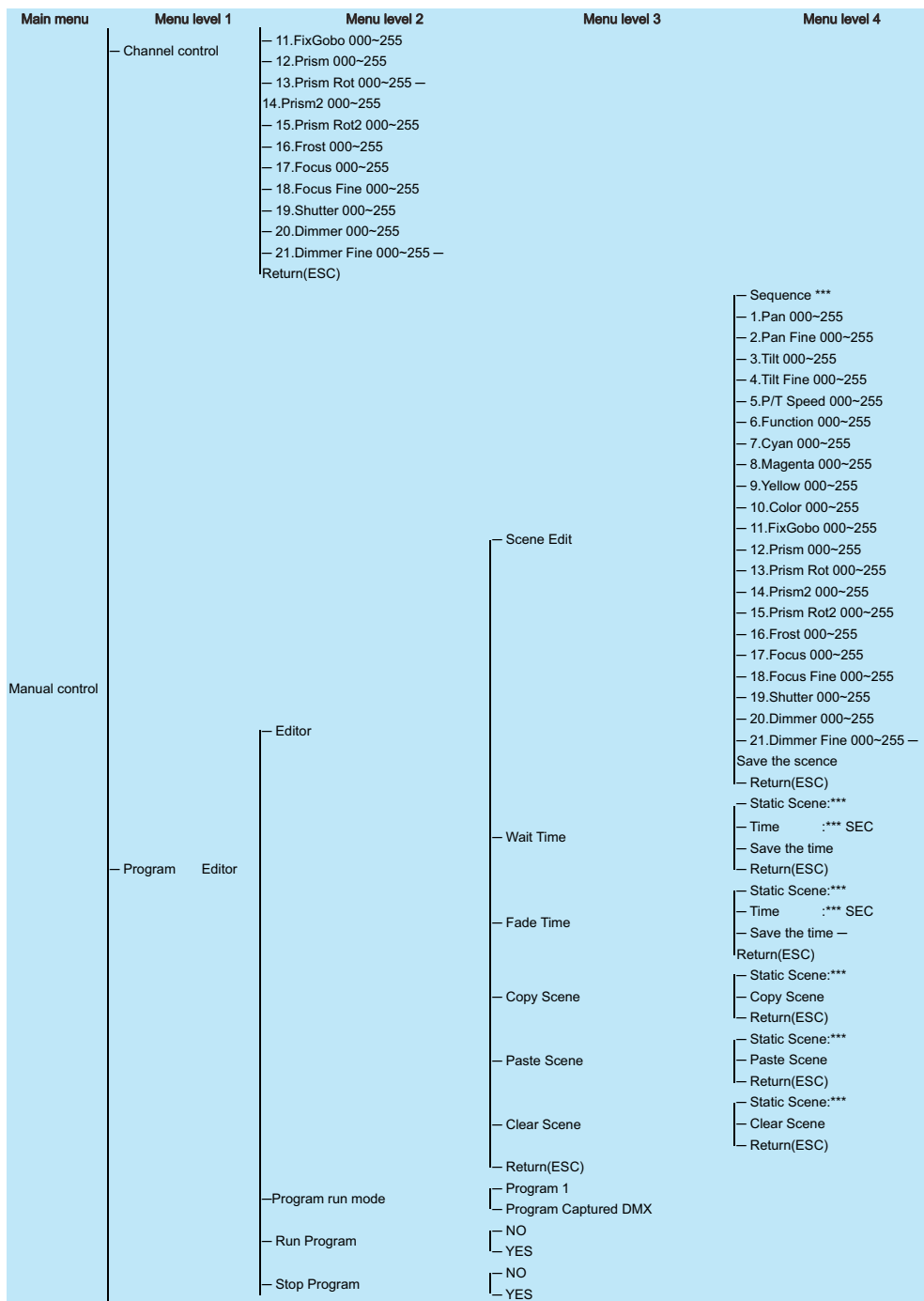
- NO DMX: Display flashes and shows at 'DMX: X'.
- DMX: Display backlight turns off and shows 'DMX: V'.
- The fixture is fully RDM ready. For RDM functions please refer to the ANSI/ESTA E1.20-2006 standard.

ONBOARD CONTROL MENU

Main menu	Menu level 1	Menu level 2	Menu level 3	Menu level 4
DMX Address	— Address:001-512	— Address(**.**) (Default : 001)		
	— DMX Signal mode	— Wire- -WDMX OFF(Default)		
		— Wireless- -WDMX ON		
		— Ethernet		
		— Control Protocol	— ArtNet	
			— sACN	
	— Ethernet	— IP Address	— ***, ***, ***, **	
		— Mask Address	— ***, ***, ***, **	
		— Artnet Net ADD	— 1-128	
		— Artnet Sub Net ADD	— 0-15	
		— Artnet Universe ADD	— 0-15	
	— Return(ESC)	— sACN Universe	— -1-63999	
		— DHCP	— OFF	
			— ON	
	— Power hours	— Total Hours: ***(H)		
		— Rst Hours: ***(H)		
	— Laser hours	— Total Hours: ***(H)		
		— Rst Hours: ***(H)		
	— Temperature	— In TEMP :000.0		
Information	— Logged temperature	— In temperature	— Cur TEMP: ***	
			— Max TEMP: ***	
		— Return(ESC)	— Min TEMP: ***	
	— Fan information	— Head fan	— 1.Power1:**.°V	
			— 2.Sp-Fan1:**.°%	
		— Focus fan	— 3.Sp-Fan1:****R	
			— 4.Sp-Fan2:**.°%	
		— Base fan	— 5.Sp-Fan2:****R	
			— 1.Power:**.°V	
	— RDM UID		— 2.Sp-Fan1:**.°%	
			— 3.Sp-Fan1:****R	
			— 4.Sp-Fan2:**.°%	
	— DMX live	— Return(ESC)	— 5.Sp-Fan2:****R	
		— 02E2:*****	— 1.Power:**.°V	
		— 1.Pan 000~255	— 2.BaseFan1:**.°%	
		— 2.Pan Fine 000~255	— 3.BaseFan1:****R	
		— 3.Tilt 000~255		
		— 4.Tilt Fine 000~255		
		— 5.P/T Speed 000~255		
		— 6.Function 000~255		
		— 7.Cyan 000~255		
		— 8.Magenta 000~255		
		— 9.Yellow 000~255		
		— 10.Color 000~255		
		— 11.FixGobo 000~255		
		— 12.Prism 000~255		
		— 13.Prism Rot 000~255		
		— 14.Prism2 000~255		
		— 15.Prism Rot2 000~255		
		— 16.Frost 000~255		
		— 17.Focus 000~255		
		— 18.Focus Fine 000~255		
		— 19.Shutter 000~255		
		— 20.Dimmer 000~255		
		— 21.Dimmer Fine 000~255		

This menu can only be displayed when the artnet module is installed

Main menu	Menu level 1	Menu level 2	Menu level 3	Menu level 4
	- System version - XY Board: V*. ** - LED Board: V*. ** - CMY Board: V*. ** - DIS Board: V*. ** - Return(ESC)			
Personal	Display lock	- OFF - ON		
	Wireless options	Wireless ON/OFF	- OFF - ON	
		wireless settings	- Idle - Unlink transmitter – Wireless off	
		wireless status	- Not found transmitt - No wireless DMX - Connected Transmitt - No wireless DMX - Connected Transmitt - Wireless DMX connect	This is an invisible menu that will only be displayed when a wireless module is inserted
	Frequency	1800Hz 3600Hz (Default) 7200Hz 25000Hz		
	Fan mode	- standard (Default) - Silent		
	Dimmer speed	- Dimmer speed normal - Dimmer speed Smooth		
	Dimmer curve	- Linear - Square (Default) - I-Square - S Curve		
	P/T invert	Pan invert	- OFF - ON	
		Tilt invert	- OFF - ON	
		Return(ESC)		
	Display	Backlight	- Always on - Auto OFF(15s)	
		Rotate	- Normal - Rotate 180	
		Backlight blink	- ON - OFF	
		Return(ESC)		
	P/T speed mode	- Normal - Fast - Precise		
	CMY TV mode	- OFF - ON		
	Return(ESC)			
		1.Pan 000~255 2.Pan Fine 000~255 3.Tilt 000~255 4.Tilt Fine 000~255 5.P/T Speed 000~255 6.Function 000~255 7.Cyan 000~255 8.Magenta 000~255 9.Yellow 000~255 10.Color 000~255		



Main menu	Menu level 1	Menu level 2	Menu level 3	Menu level 4
	System reset	Run on power on Capture DMX Return(ESC) Reset all Pan/Tilt reset Head motor reset Return(ESC)	NO YES Saved DMX data Return(ESC)	Static Scene:*** Saved Scene from DMX – Return(ESC)
	Return(ESC)			
Service	Error information	Error list Empty list Return(ESC)	NO OFF	
	Reset Timers	Reset Power Hours Reset Laser Hours Return(ESC)	NO YES NO YES	
		Load default	Go back Load Default	
		Reset Total Timers	Total Power Hours Total Laser Hours Return(ESC)	
		Clear logged temperature	-Password required	
	Factory			Pan 000-255 Tilt 000-255 Cyan 000-255 Magenta 000-255 Yellow 000-255 Color 000-255 FixGobo 000-255 Prism1 000-255 Prism Rot 000-255 Prism2 000-255 Frost 000-255 Focus 000-255 Dimmer 000-255 Return(ESC) 000-255
		Calibration	-Password required	
		Developer	-Password required	
	Return(ESC)	Return(ESC)		
Language				
Test	Test P/T	STEP ***		
	Test effect	STEP ***		
	Test all	STEP ***		
	Return(ESC)			
Rotate display				
	Normal			
	Rotate 180			

DMX PROTOCOL

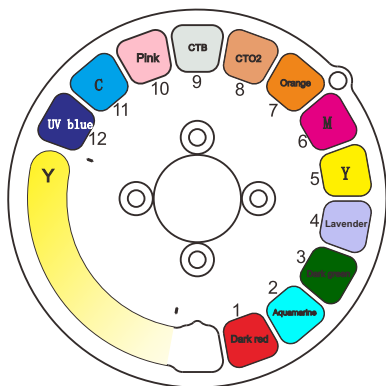
CHANNEL	Function	VALUE	SETTING	NOTE
1	Pan	0-255	Pan positioning	
2	Fine Pan	0-255	Fine Pan positioning	Angle : 0-540° Maximal Speed : 2.11S
3	Tilt	0-255	Tilt positioning	
4	Fine Tilt	0-255	Fine Tilt positioning	Angle : 0-270° Maximal Speed:1.15S
5	Speed pan/tilt	0-255	Speed pan/tilt	
6	Device Set (Hold 3s)	0-9	Reserved	
		10-14	Display lock off	
		15-19	Display lock on	
		20-24	Display rotate normal	
		25-29	Display rotate 180°	
		30-34	Display backlight always on	
		35-39	Display backlight auto off(15s)	
		40-44	Display backlight blink off	
		45-49	Display backlight blink on	
		50-54	XYNormal mode	
		55-59	XYFast mode	
		60-64	XYPrecise mode	
		65-69	Reserved	
		70-74	CMYTVmode Off	
		75-79	CMYTVmode on	
		80-84	Reserved	
		85-89	Reserved	
		90-94	Reserved	
		95-99	Reserved	
		100-104	Pan invert off	
		105-109	Pan invert on	
		110-114	Tilt invert off	
		115-119	Tilt invert on	
		120-124	Reserved	
		125-129	Reserved	
		130-139	Reserved	
		140-149	Pan/Tilt reset	
		150-159	Head motor reset	
		160-169	Total reset	
		170-174	Fan Standard mode	
		175-179	Fan Silent mode	
		180-184	LED frequency 1.8kHz	
		185-189	LED frequency 3.6kHz	
		190-194	LED frequency 7.2kHz	
		195-199	LED frequency 25kHz	
		200-204	Dimmer curve linear	
		205-209	Dimmer curve square	
		210-214	Dimmer curve inv-square	
		215-219	Dimmer curve "S"	
		220-224	Dimmer speed normal	
		225-229	Dimmer speed smooth	
		230-234	Reserved	
		235-239	Reserved	
		240-255	Reserved	

CHANNEL	Function	VALUE	SETTING	NOTE
7	Cyan		Cyan	
8	Magent		Magenta	
9	a Yellow		Yellow	
10		0-4	White	
		5-8	White + Dark red	
		9-12	Dark red	
		13-17	Dark red +	
		18-21	Aquamarine	
		22-25	Aquamarine + Dark green	
		26-29	Dark green	
		30-34	Dark green + Lavender	
		35-38	Lavender	
		39-42	Lavender + Yellow	
		43-46	Yellow	
		47-51	Yellow + Magenta	
		52-55	Magenta	
		56-59	Magenta + Orange	
		60-63	Orange	
		64-68	Orange+CTO2 (2500K)	
		69-72	CTO2 (2500K)	
		73-76	CTO2 (2500K) + CTB (8000K)	
		77-81	CTB (8000K)	
		82-85	CTB (8000K) +	
		86-89	Pink Pink	
		90-93	Pink + Cyan	
		94-98	Cyan	
		99-102	Cyan + Dark blue	
		103-106	Dark blue	
		107-127	Reserved	
		128-191	color wheel shake fast to slow	5-0.2HZ
		192-255	color wheel shake slow to fast	0.2-5HZ
11	Fixgobo Wheel	0-3	Open	
		4-8	FixGobo 1	
		9-13	FixGobo 2	
		14-18	FixGobo 3	
		19-23	FixGobo 4	
		24-28	FixGobo 5	
		29-33	FixGobo6	
		34-38	FixGobo 7	
		39-43	FixGobo 8	
		44-48	FixGobo 9	
		49-53	FixGobo 10	
		54-58	FixGobo 11	
		59-63	FixGobo 12	
		64-68	FixGobo 13	
		69-73	FixGobo 14	
		74-78	FixGobo 15	
		79-83	FixGobo 16	
		84-88	FixGobo 17	
		89-95	FixGobo1shake , slowtofast	Speed : 0.2->10 HZ
		96-102	FixGobo2shake , slowtofast	Speed : 0.2->10 HZ
		103-109	FixGobo3shake , slowtofast	Speed : 0.2->10 HZ
		110-116	FixGobo4shake , slowtofast	Speed : 0.2->10 HZ

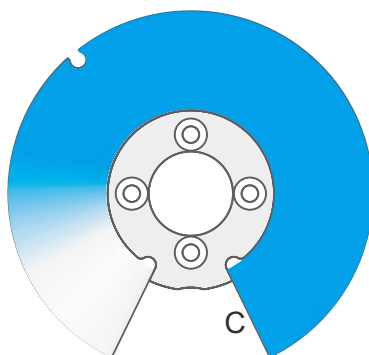
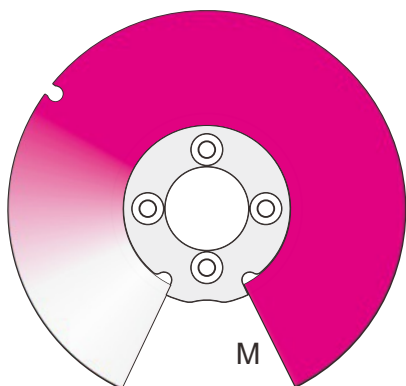
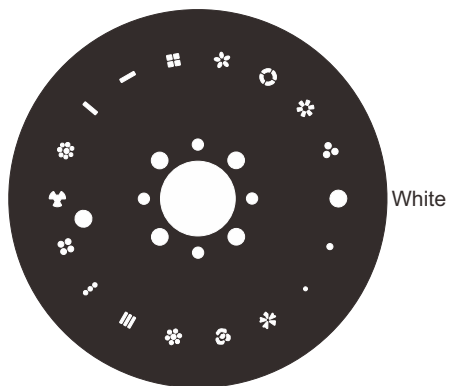
CHANNEL	Function	VALUE	SETTING	NOTE
		117-123	FixGobo5 shake , slow to fast	Speed : 0.2→10 HZ
		124-130	FixGobo6 shake , slow to fast	Speed : 0.2→10 HZ
		131-137	FixGobo7 shake , slow to fast	Speed : 0.2→10 HZ
		138-144	FixGobo8 shake , slow to fast	Speed : 0.2→10 HZ
		145-151	FixGobo9 shake , slow to fast	Speed : 0.2→10 HZ
		152-158	FixGobo10 shake , slow to fast	Speed : 0.2→10 HZ
		159-165	FixGobo11 shake , slow to fast	Speed : 0.2→10 HZ
		166-172	FixGobo12 shake , slow to fast	Speed : 0.2→10 HZ
		173-179	FixGobo13 shake , slow to fast	Speed : 0.2→10 HZ
		180-186	FixGobo14 shake , slow to fast	Speed : 0.2→10 HZ
		187-193	FixGobo15 shake , slow to fast	Speed : 0.2→10 HZ
		194-200	FixGobo16 shake , slow to fast	Speed : 0.2→10 HZ
		201-207	FixGobo17 shake , slow to fast	Speed : 0.2→10 HZ
		208-227	FixGobo Wheel CW fast to slow	Speed : 120→0 RPM
		228-229	Stop	
		230-249	FixGobo Wheel CCW slow to fast	Speed : 0→120 RPM
		250-255	Auto radom gobo selection from fast to slow	
12	Prism	0-3	Prism out	
		4-199	Prism in	
		200-255	Shaking Prism from slow to fast	
13	PrismRot	0-63	Prism indexing	Angle : 0-360°
		64-127	Forwards prism rotation from fast to stop	Speed : 62.5→0 RPM
		128-191	Backwards prism rotation from stop to fast	Speed : 0→62.5 RPM
		192-207	from slow to fast 90° Swing	0.2→1 HZ
		208-223	from slow to fast 180° Swing	0.2→1 HZ
		224-239	from slow to fast 270° Swing	0.2→1 HZ
		240-255	from slow to fast 360° Swing	0.2→1 HZ
14	Prism2	0-3	Prism out	
		4-199	Prism in	
		200-255	Shaking Prism from slow to fast	
15	PrismRot2	0-255	Reserved	
16	Frost	0-3	Frost Out	
		4-255	Frost In	
17	Focus	0-255	Focus 8bit	
18	FocusFine	0-255	FocusFine 16bit	
		0-31	Strobe closed	
		32-63	Strobe open	
	96 - 127	64-95	Synchronous strobe,slow → fast	1-25HZ
	128 - 143	96-127	Strobe open	
	144 - 159	128-143	Fast closing , Slow opening , Slow → Fast	0.5-12HZ
	160 - 191	144-159	Slow closing , Fast opening , Fast → Slow	12-0.5HZ
	192 - 223	160-191	Strobe open	
	224 - 255	192-223	Random strobe,Slow → Fast	1-25HZ
		224-255	Strobe open	
20	Dimmer	0-255	Dimmer 0% to 100% (0=default)	
21	Dimmer Fine	0-255	Dimmer Fine 0% to 100% (0=default)	

GOBO OVERVIEW

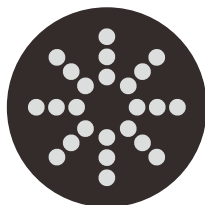
Color wheel



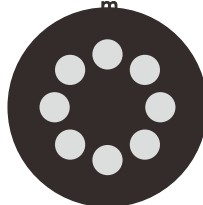
Static gobo wheel



24 Prism



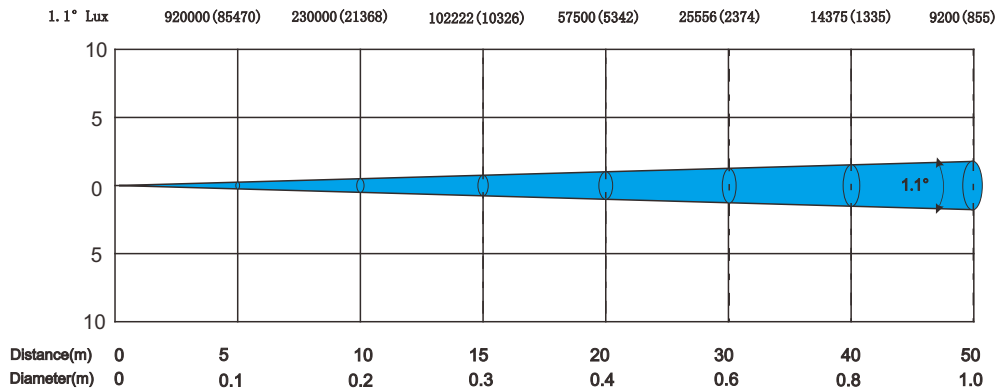
8 Prism



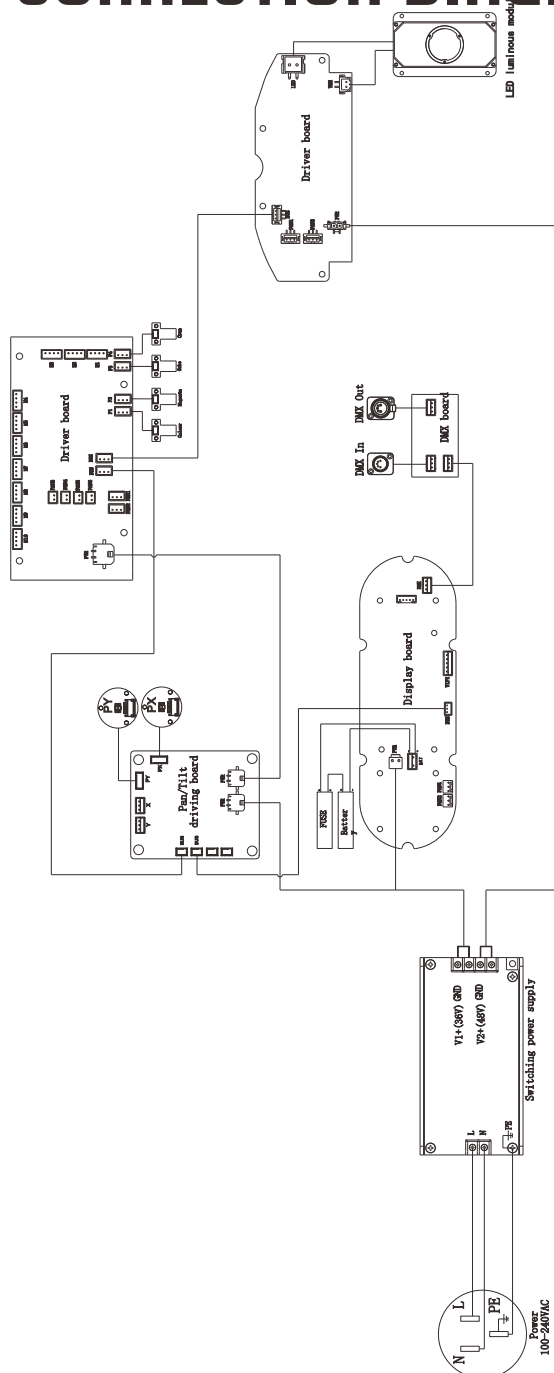
PHOTOMETRICS

Distance, spot diameter and illumination diagram

Lux (fc)



CIRCUIT CONNECTION DIAGRAM



SPECIFICATIONS

Power

Input voltage & rate	AC100-240V~ 50/60Hz
Standby power	68W
Total power consumption(at nominal voltage 230V)	233W
Typical current (at nominal voltage 230V)	2.33A
Cos ϕ -	0.996
Power plug type	Seetronic Powercon TRUE 1

Configuration

Light Source	SH-LS120W (Laser)
Expected Lifetime	12000 hours (under best possible conditions)
Color temperature	9256 K
CRI level	70 (Standard)
Light Output	3600 lm
Dimming frequency	1800Hz/3600Hz/7200Hz/25000Hz
Dimmer resolution	65535 (16Bit)

Optical

Beam angle	1.1°
------------	------

Photometric

Output @10M	230000 lux
Output @50M	9200 lux

Effects

Color Mixing	CMY
Color wheel	Fixed color wheel 12 + open
Static Gobe wheel	Fixed wheel 17 + open
Prism wheel	8 face prism + 24 face prism
Focus	16 bit
Frost	10°
Dimmer	16 bit 0 – 100%
Shutter	0.5-25Hz / second
Pan	540°
Tilt	270°

Heat management

Cooling type	Forced ventilation with axial fans. Automatically adjust cooling-fan's speed
MAX ambient temp (Ta max)	Ta max= 45 °C
MIN ambient temp (Ta min)	Ta min= -20 °C
MAX housing temp. (ta=25°C)	Tc= 50 °C
MAX housing temp. (ta=40°C)	Tc= 60 °C

Menu

Auto program	Manual program and Auto test program
Static color	White
Manual calibration	Service → calibration → function (0-255)
Factory calibration	Yes
Strobe speed	0.5-25 times/second
Random strobe	0.5-25 times/second

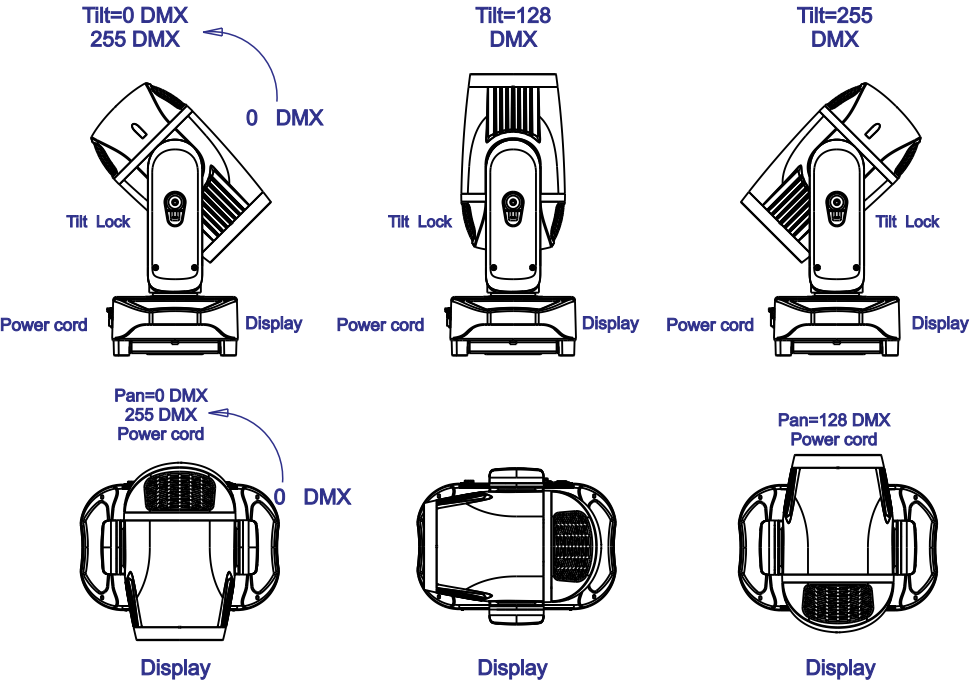
Control

Control protocol	DMX512
DMX channel range	21 channel

RDM

RDM compliance	DMX512-A
WDMX	No (optional)
ACN	No (optional)
DMX input connection	5-pin
Data input (artnet, SACN)	No
Hardware Interface	
	LCD Display
Software upload method	Software upload method XLR, firmware upload tool
Installation	
IP rating	IP65(Use rubber power & data cover)
Orientation	Any
Housing	
Safety attachment point	Bottom
Lock	Tilt lock
Physical	
Net product weight	16.1 Kg
Machine dimensions - length	364 mm
Machine dimensions - depth	241 mm
Machine dimensions - height	517 mm
Carton size	425 x 290 x 640 mm
Gross weight	19.1 Kg
Accessories	
Included items	Manual, Power cable, safety cable
Approvals	
Approved certifications	CE and RoHS
Information	
Article number	160060

POSITIONING



POSEIDON LASER BEAM XS